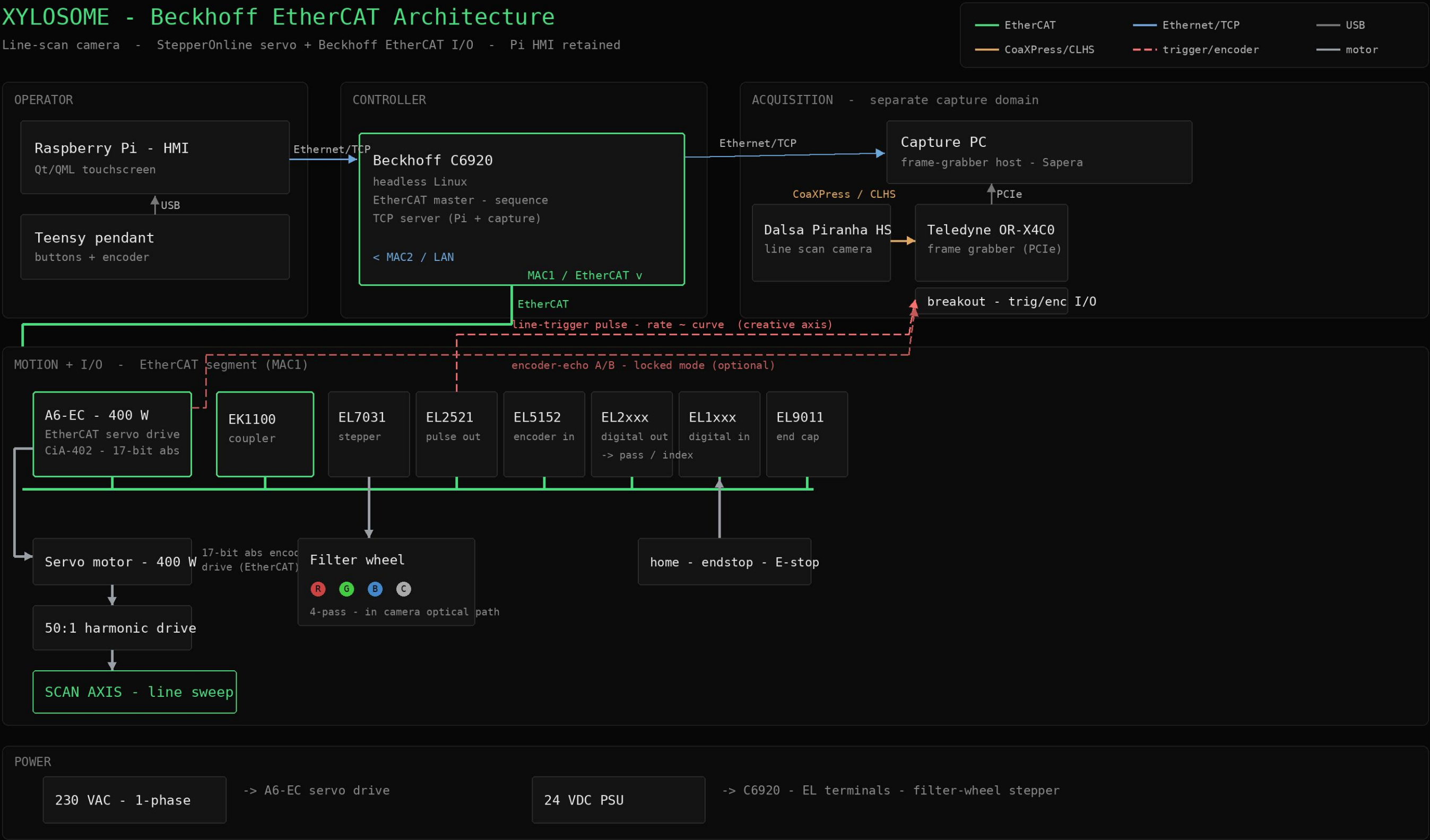


XYLOSOME - Beckhoff EtherCAT Architecture

Line-scan camera - StepperOnline servo + Beckhoff EtherCAT I/O - Pi HMI retained



THREE CONTROL AXES

- 1 - motion speed - speed curve drives the servo
- 2 - line rate - EL2521 pulse mapped to the curve
geometry <-> sampling : the new creative axis
- 3 - exposure - set on camera, independent -> brightness

OPEN ITEMS / TO VERIFY

- gearhead <-> servo flange - fit / adapter plate
- smooth CiA-402 - PREEMPT_RT or LinuxCNC-ethercat
- A6-EC encoder-echo (A/B) output - for locked mode
- capture PC hosts frame grabber (PCIe) - stays separate